

Daejeon, Korea
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Hyunmo Sung

<https://programelot.github.io>
GitHub: programelot
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SKILLS

Programming	C, C++, C#, CUDA, Python, PAPI, CMake, LLVM
Communication	Korean (native), English
Other	Unity, Visual studio code, Github, Linux(Ubuntu)

WORK EXPERIENCE

Engineer <i>Samsung Electronics</i>	March 2026 - <i>Suwon, Korea</i>
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Researcher <i>ELC(Embedded Systems Languages and Compilers Lab)</i>	February 2020 - February 2023 <i>Yonsei University, Seoul, Korea</i>
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- Researched about the profile-guided-optimization.
- Implemented and evaluated the performance of the bloom filter using CUDA unified memory.
- Evaluated the performance of lazy parallel kronecker algebra on the modern GPU T4.
- Joint Research Project with DS Division of Samsung Inc. through Yonsei-Samsung Semiconductor Research Center (YSSRC) Program.

Internship <i>ELC(Embedded Systems Languages and Compilers Lab)</i>	July 2019 - February 2020 <i>Yonsei University, Seoul, Korea</i>
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- Researched about the kronecker algebra to detect the deadlock of the program.

Internship <i>RiseGroup</i>	December 2014 - February 2015 <i>Dongguk University, Seoul, Korea</i>
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- Worked as a subprogrammer. Developed a serious game to prevent school violence.

PROJECT

Revisit the Supernodal Floyd Warshall algorithm <i>Side project</i>	November 2023 - December 2023
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- Implemented the Supernodal Floyd-Warshall algorithm from the paper <https://dl.acm.org/doi/10.1145/3332466.3374533>.
- 78 times faster than a Floyd-Warshall algorithm with a single thread.
- 17 times faster than a Floyd-Warshall algorithm with multiple threads with other optimization techniques.
- Utilized entire threads with OpenMP.
- Source is open on the github repository <https://github.com/programelot/Supernodal-Floyd-Warshall>.

Matrix multiplication on GPU <i>Side project</i>	July 2022 - September 2022
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- Developed 11 matrix multiplication algorithms on CPU and GPU.
- Estimated and compared performance of each algorithms.
- Utilized the shared memory on GPU.
- Implemented Strassen's algorithm and Winograd's algorithm.
- Source is open on the github repository <https://github.com/programelot/MatrixMultiplication>.

BackRomii

Indie Game Development

July 2022

- Developed an escape game from auto generated maze with first person view.
- - Used recursive division method to generate maze.
- Optimized the game by developing an off culling algorithm.
- Developed a path finding algorithm.
- Developed in two weeks like game jam.
- Game has been released on the web. <https://aintmos.itch.io/backromii>
- 1162 views and 292 downloads happened.

EDUCATION

Computer science

March 2020 - February 2023

Yonsei university, Seoul, Korea

Master, Graduated

Main courses: Multicore computing topics, Introduction to approximation algorithms

Computer science

March 2016 - February 2020

Yonsei university, Seoul, Korea

Bachelor, Graduated

Main courses: Algorithm analysis, Computer graphics, Compiler design, Multicore programming fundamentals

Multimedia engineering

March 2014 - February 2016

Dongguk university, Seoul, Korea

Bachelor, Drop out for transfer to other university

Main courses: Multimedia Data Structures, Internet Programming, Multimedia Programming

PUBLICATION

Profile-guided Orchestration of Computations for CPU and PIM Cores

2023

Hyunmo Sung

Yonsei university, thesis.

Performance Evaluation of GPU-based Bloom Filters Using CUDA Unified Memory

2022

Hyunmo Sung, and Bernd Burgstaller

Korea Software Congress 2022 (한국정보과학회 학술발표논문집 2022): 45-47.

Lazy Evaluation of Kronecker Algebra Operations on the Tesla T4 GPU

2020

Ham, Seokhwan, **Hyunmo Sung**, Shinyung Yang, and Bernd Burgstaller

Korea Computer Congress 2020 (한국정보과학회 학술발표논문집 2020): 44-46.

PATENT

Offloading methodology for utilizing Processing-In-Memory and the machine for it

(프로세스 인 메모리의 활용을 위한 오프로드 처리 방법 및 그를 위한 장치)

Bernd Burgstaller, **Hyunmo Sung**, Seongho Jeong, Shinyung Yang, Jayhwan Lee, and Jiun Jung.

Application No. 10-2022-0162906, Nov 29 2022.

AWARDS

- Capstone project 1st place (졸업 작품 최우수상)

December 06, 2019

Lazy Parallel Kronecker Algebra, Yonsei University, Seoul, Korea

- **Capstone project 1st place (졸업 작품 최우수상)** **May 13, 2019**
Projection-Based AR Evacuation Simulator using Kinect for Windows V2, Yonsei University, Seoul, Korea
- **Honored Student Prize (학기 우등생)** **July 09, 2015**
Dongguk University, Seoul, Korea
- **Honored Student Prize (학기 우등생)** **January 09, 2015**
Dongguk University, Seoul, Korea
- **Honored Student Prize (학기 우등생)** **July 07, 2014**
Dongguk University, Seoul, Korea